CHAPTER 9

POPULATION: DISTRIBUTION, DENSITY, GROWTH AND COMPOSITION

CHAPTER ESSENCE AT A GLANCE:
Master India demographic patterns: spatial population distribution, arithmetic versus physiological density, the four phases of demographic growth since 1901, the Year of the Great Divide, linguistic groups, and occupational categories.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Spatial Distribution and Density:** India population is highly clustered; Uttar Pradesh is the most populous state, while Bihar has the highest density (1,106 persons per sq km). Physiological density and agricultural density measure pressure on arable land far more accurately than arithmetic density.

• **Four Historical Phases of Growth:** Phase 1 (1901 to 1921): Stagnant stationary phase. Phase 2 (1921 to 1951): Steady growth (1921 is the Year of the Great Divide). Phase 3 (1951 to 1981): Population explosion with rapid drop in mortality. Phase 4 (1981 to 2011): High growth with progressive deceleration in birth rates.

• **Linguistic and Religious Diversity:** Speakers belong to four major language families: Indo-Aryan (73 percent), Dravidian (20 percent), Austro-Asiatic or Nishada (1.38 percent), and Sino-Tibetan or Kirata (0.85 percent). Major religions include Hinduism, Islam, Christianity, Sikhism, Buddhism, and Jainism.

• **Occupational Structure:** The Census classifies population into Main Workers (working 183 days or more in a year), Marginal Workers (working less than 183 days), and Non-Workers. The workforce shows a progressive structural shift from primary to tertiary sectors.

ESSENTIAL GEOGRAPHICAL MODELS & LANDFORMS

$$\text{Physiological Density} = \frac{\text{Total Human Population}}{\text{Net Cultivated Agricultural Area}}$$

Physiological Density on Cultivated Land

$$\text{Agricultural Density} = \frac{\text{Total Agricultural Population}}{\text{Net Cultivated Agricultural Area}}$$

Agricultural Labour Density Ratio

$$\text{Decadal Population Growth Rate } (\%) = \frac{P_2 - P_1}{P_1} \times 100$$

Inter-Censal Decadal Percentage Growth

$$\text{Sex Ratio (India)} = \frac{\text{Number of Females}}{\text{Number of Males}} \times 1000$$

Female to Male Sex Ratio Formula

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** The year 1921 is designated as the 'Year of the Great Divide' because prior to 1921 population growth was erratic and experienced negative growth (1911 to 1921), after which growth was consistently positive.

• **Alert:** Physiological density divides total human population by net cultivated area, whereas agricultural density divides strictly the agricultural population by net cultivated area.

• **Alert:** To be classified as a Main Worker by the Census of India, a person must have engaged in economically productive work for at least 183 days (six months) during the preceding year.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Explain the difference between physiological density and agricultural density of population. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Physiological Density:** Calculated by dividing the total population of a country by the net cultivated agricultural area (Total Population divided by Net Cultivated Area). **[1.5 Marks]**

• **Agricultural Density:** Calculated by dividing the total agricultural population (cultivators and agricultural labourers) by the net cultivated area (Agricultural Population divided by Net Cultivated Area). **[1.5 Marks]**

2. Describe the four distinct phases of population growth in India since 1901. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **Phase 1 (1901 to 1921):** Period of stagnant or stationary growth; both birth rate and death rate were high; 1911 to 1921 recorded a negative growth rate (-0.31%) due to famines and influenza. **[1 Mark]**

• **Phase 2 (1921 to 1951):** Period of steady population growth; improvements in overall sanitation and healthcare reduced mortality, while crude birth rate remained elevated. **[1 Mark]**

• **Phase 3 (1951 to 1981):** Period of population explosion; rapid drop in mortality due to centralized developmental planning and modern antibiotics, causing very high natural increase. **[1 Mark]**

• **Phase 4 (1981 to 2011):** High growth rate with definite signs of slowing down; downward trend in fertility due to female education, rising marriage age, and family welfare programmes. **[1 Mark]**

3. Why is the year 1921 celebrated as the 'Year of the Great Divide' in India demographic history? [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Erratic Past:** Prior to 1921, India population growth was irregular, fluctuating, and actually recorded negative growth between 1911 and 1921 (-0.31 percent). **[1 Mark]**

• **Unbroken Expansion:** From 1921 onwards, mortality was brought under control and population grew steadily without experiencing a single year of demographic decline. **[1 Mark]**

4. Name the four major language families of India and state their approximate demographic shares. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **1. Indo-Aryan (Indo-European):** Largest family, spoken by about 73 percent of the total population across northern and central India. **[1 Mark]**

• **2. Dravidian:** Second largest family, spoken by approximately 20 percent of the population across southern peninsular India. **[1 Mark]**

• **3. Austro-Asiatic (Nishada) and Sino-Tibetan (Kirata):** Austro-Asiatic accounts for 1.38 percent, and Sino-Tibetan accounts for 0.85 percent in tribal and Himalayan tracts. **[1 Mark]**

5. Distinguish between main workers and marginal workers according to the Census of India. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Main Worker:** A person who participates in economically productive work for at least 183 days (or six months) during the reference year. **[1.5 Marks]**

• **Marginal Worker:** A person who participates in economically productive work for less than 183 days (or less than six months) during the reference year. **[1.5 Marks]**

6. Discuss the spatial patterns of population density across Indian states, highlighting regions of very high and very low density. **[Score: 3 Marks]**

Answer & Key Valuation Steps:

• **Very High Density:** Northern plains and coastal states have very high densities due to fertile alluvium, flat relief, and abundant water (Bihar with 1,106, West Bengal with 1,028, and Kerala with 860 persons per sq km). **[1.5 Marks]**

• **Very Low Density:** Rugged mountainous terrains and arid states exhibit low densities due to harsh environmental constraints (Arunachal Pradesh with only 17 persons per sq km, Mizoram, and Sikkim). **[1.5 Marks]**

7. Explain the occupational structure of India population and highlight the sectoral shifts observed in recent decades. **[Score: 3 Marks]**

Answer & Key Valuation Steps:

• **Three Broad Sectors:** Workforce is divided into primary activities (cultivators, agricultural labourers), secondary activities (manufacturing, household industry), and tertiary activities (trade, transport, services). **[1 Mark]**

• **Primary Sector Contraction:** The proportion of workers dependent on agriculture has steadily declined, dropping below 55 percent. **[1 Mark]**

• **Secondary and Tertiary Expansion:** Employment in service sectors, construction, retail, and manufacturing has expanded steadily, indicating structural modernization. **[1 Mark]**